

MATH 1342 – Elementary Statistics – Week 8: Percentile Intervals and Introducing Hypothesis Tests

Oct 13 – 15 | 3.4 Bootstrap Confidence Intervals Using Percentiles • 4.1 Introducing Hypothesis Tests | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
3. Find and use empirical probabilities in bootstrap distributions to find confidence intervals and in randomization distributions to test hypotheses. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.	1. Determine which percentiles of a bootstrap distribution bound a 90%, 95%, or 99% confidence interval, and construct the interval with StatKey. (CO3, CO9) 2. Predict how changing the confidence level or the sample size affects the width of an interval. (CO9) 3. Judge whether a bootstrap confidence interval is trustworthy from the shape of the bootstrap distribution and whether the sample is representative. (CO3) 4. Identify the parameter of interest and write null and alternative hypotheses from a research question. (CO10) 5. Classify a test as right-tailed, left-tailed, or two-tailed, and explain the logic of assuming H_0 and asking whether the data are surprising. (CO10)	e-Text: 3.4 Bootstrap Confidence Intervals Using Percentiles	MO1, MO2, MO3	Participation: Percentile Confidence Intervals – StatKey: 90/95/99% intervals for the same data; compare SE and percentile methods	MO1, MO2, MO3	Online Homework: 3.4 Bootstrap Confidence Intervals Using Percentiles (due Sun Oct 18, 11:59 PM)	MO1, MO2, MO3
		Interactive Lecture: 3.4 Bootstrap Confidence Intervals Using Percentiles	MO1, MO2, MO3	Participation: Writing Hypotheses – Zener-card ESP practice (26 of 50): define the parameter, state H_0 and H_a , choose the tail	MO4, MO5	Online Homework: 4.1 Introducing Hypothesis Tests (due Sun Oct 18, 11:59 PM)	MO4, MO5
		Slides: 3.4 Bootstrap Confidence Intervals Using Percentiles	MO1, MO2, MO3	Think about it: is 15 of 16 correct (dolphin study) convincing?	MO5	Take-Home Quiz 4 (due Tue Oct 13, 11:59 PM) – covers P.1, 3.1–3.3	None
		e-Text: 4.1 Introducing Hypothesis Tests	MO4, MO5				
		Interactive Lecture: 4.1 Introducing Hypothesis Tests	MO4, MO5				
		Slides: 4.1 Introducing Hypothesis Tests	MO4, MO5				